Smith, "Closing the Loop" April 15, 2026

- How has Newton's theory of gravity been tested over the three centuries since the *Principia*? Smith characterizes an alternative account of theory-testing, differs profoundly from the standard hypothetico-deductive picture (derive predictions, compare with observation), and argues that it provides much more compelling evidence.
- What kind of "knowledge" survives the transition from Newtonian to Einsteinian gravity? If the most successful theory in the history of science was ultimately replaced, what claim to permanence can any scientific knowledge have?
- How does the history of gravity research illuminate the role of theory as an *instrument of ongoing research* — not an explanation or representation of known phenomena, but a tool for discovering new things about the world?

I. The Evidence Problem and Newton's Response

Smith begins from the observation that Newton himself recognized, as he was writing the *Principia*, that the true orbital motions are inordinately complicated — each planet traces a fresh orbit every revolution, due to interactions with all the other planets. The complexity of the true motions means there will always be room for competing theories that agree with observation to any given level of approximation. (This is essentially one of the problems that Duhem later recognized.) On Smith's account: the *Principia* is "one sustained response to this evidence problem."

Phenomena and *Quam Proxime* Reasoning. Newton's "phenomena" are not precise observational facts; they are descriptions of regularities that hold at least *quam proxime* ("very, very nearly") over a finite body of observations, from a limited period of time. Newton takes care to establish that his key enabling "if-then" propositions hold in the *quam proxime* form as well:

> If a body sweeps out equal areas in equal times *quam proxime* with respect to some body T, then the force governing its motion is directed *quam proxime* toward T, and conversely.
>
> If the apsidal angle for a nearly circular orbit is approximately θ then the force law is approximately $f \propto r^{n-3}$, where $n = \left(\frac{\theta}{\pi}\right)^2$ (Prop I.45).

What Newton can strictly "deduce" from phenomena is limited: that his law of gravity held to high approximation over a particular period of time. The inference to stronger claims (that the law is exact and universal) is governed by Rules of Reasoning: instructions for how to proceed in continuing research, authorizing claims to be *provisionally* established until further phenomena give reasons to reconsider.

Newtonian Idealizations. Taking the theory to be exact has a specific methodological payoff: it yields claims about the world that *would hold exactly in certain specifiable circumstances.* Smith calls these "Newtonian idealizations" — approximations that, according to theory, would hold exactly under specific conditions (including provisos of no further forces or density variations). Examples:

- The area rule would hold exactly in the absence of forces from other orbiting bodies.
- Orbits would be perfectly stationary Keplerian ellipses if no perturbing forces were acting.
- Gravity around a perfect, spherically symmetric body would vary exactly as the inverse-square of distance from the center.

Newton knew these are all counterfactual claims: the conditions are never actually realized. But taking the theory as exact allows the *contrapositive*: every systematic discrepancy between observation and the idealized result must stem from a physical source not taken into consideration.

Second-Order Phenomena. The discrepancies between Newtonian idealizations and observation are what Smith calls "second-order phenomena": systematic patterns that are not directly observable, but emerge through the comparison between idealized calculations and observations. They are "second-order" because they presuppose the theory of gravity, taken to hold exactly; they are "phenomena" because they are systematic regularities, often initially masked by more dominant regularities.

II. The Logic of Theory-Testing: "Closing the Loop"

Smith argues that the test question forced on gravity research following the *Principia* was not whether calculation agrees with observation, but *whether robust physical sources can be found for discrepancies between calculation and observation.* Persistent failure to identify robust sources for discrepancies would indicate the need to treat Newtonian gravity as *not* exact, or *liable to exceptions.*

Two forms of testing:

- **Basic form:** Pin down sources of discrepancies, confirm they are robust and physically significant, while progressing toward smaller and smaller discrepancies.
- **Ramified form:** Previously discovered physical sources of second-order phenomena become incorporated into further calculations, so that new second-order phenomena presuppose them as well as the theory of gravity. This makes subsequent testing *ever more stringent* by constraining the freedom to find sources of newly emerging, still smaller deviations.

The Feedback Loop (Figure 6 of the essay). The general form of the evidence:

1. Start with a calculation of orbital motion that gravity theory entails would hold exactly if no other forces were at work.
2. Comparison with observation yields a discrepancy with a clear signature — a second-order phenomenon.
3. Find a robust physical source for this discrepancy, compatible with the theory and with prior calculations.
4. Incorporate the source (and all its further consequences) into a new, more refined calculation: *closing the loop*, and reinitiating the process.

Evidence accrues at three points in this process: (i) a discrepancy with a clear signature (as opposed to merely random deviations) is evidence that the idealized calculation is in some respect physically correct; (ii) finding a robust physical source compatible with the theory is evidence *for* the theory, due to its essential role in discovering the further source; (iii) closing the loop, yielding still smaller discrepancies and increasingly subtle, more constrained sources, provides evidence for both the theory and all previously identified sources. (This all assumes that the closing the loop strategy involves refinements that are compatible with the basic assumptions of the theory.)

Why this is stronger than hypothetico-deductive testing: The process shows that the theory amounts to more than a curve-fit or a mathematical representation that happens to agree with observation. A sequence of successes in pinning down robust physical sources for increasingly subtle second-order phenomena, under increasingly stringent constraints, is evidence that the theory really does capture something

about the physical world. If it did not, discrepancies should emerge for which no robust physical source can be found consistent with the theory.

III. Historical Examples

Neptune. The famous case of discovering a new planet from second-order phenomena in the motion of Uranus. Smith emphasizes that Neptune is *atypical* — it remains the sole example of a new planet discovered this way — but valuable as a simple illustration of the logic. Also illustrative of the problem of underdetermination: the anomaly in Uranus's motion was not sufficient to determine both the mass and distance of the hypothesized planet; the orbits proposed by Leverrier and Adams were not that accurate, but enabled discovery nonetheless.

The Great Inequality of Jupiter and Saturn. An anomalous, long-period fluctuation in the motions of Jupiter and Saturn — nearly 900 years in period, peaking at 60 arc-minutes for Saturn. The physical source (discovered by Laplace, 1785): higher-order perturbations arising from the interaction of the two planets' orbital eccentricities and the relative locations of the centers of their elliptical orbits.

- The discovery did not involve finding something new in the world (like Neptune); the eccentricities and orientations of the orbits were already known. What was discovered was that these comparatively subtle details were *making a difference* that had gone unnoticed.
- *This is a more typical case*: what is generally discovered is that certain details among the indefinitely many in the world are making a particular detectable difference that those details had not been recognized as making.

Spencer Jones and the Fluctuation in Earth's Rotation. Smith's extended, central example of how the closing-the-loop logic produces extraordinarily strong evidence. The Hill-Brown theory of the Moon (1919) contained 1400+ perturbational terms derived from gravity, plus one "Great Empirical Term" needed to bring calculations into agreement with observation. The research question: is its source a fluctuation in the Earth's rotation, or some further unaccounted-for force on the Moon?

- Spencer Jones (1939): if the source is a fluctuation in the Earth's rotation (a systematic error in sidereal time), then Mercury, Venus, and the Sun should each display a residual discrepancy with the *same period* as the Moon's Great Empirical Term, but with a magnitude scaled by the ratio of the mean daily motions.
- He took Newcomb's theory of Mercury (with the perihelion corrected for general relativity), derived residual discrepancies from observation, renormalized by the ratio of the Moon's to Mercury's mean daily motion, and showed a striking correlation with the Moon's residual discrepancy. Same for Venus and the Sun (i.e., the Earth).
- The strength of this evidence: the clear signature in the Moon's residuals emerged only because all 1400+ perturbational terms of the Hill-Brown theory had been correctly accounted for — failure to include any one of a large fraction of them would have masked or distorted the signal. Similarly for Mercury. Hundreds of details that make a difference entered into the calculation.

A philosophically important twist: the evidence that the Hill-Brown theory provided for gravitational theory became *stronger thanks to the need for the Great Empirical Term* than it would have been had the theory simply matched observations without it. "The exception really does 'prove' the rule when robust physical sources are found for it."

IV. The Perihelion of Mercury and the Transition to Einstein

The anomaly in the precession of Mercury's perihelion: 43 arc-seconds per century, announced by Leverrier (1859), confirmed by Newcomb (1882, 1895). For 56 years, this was treated as just another discrepancy for which a physical source had yet to be found. Numerous proposed sources were rejected — a planet "Vulcan" inside Mercury's orbit, a non-symmetric mass distribution in the Sun, a small change in the exponent of r in the law of gravity. Constraints from the orbits of the other inner planets, and from the motion of the lunar apogee (Brown, 1903), made it extremely difficult to find a physical source compatible with Newtonian theory.

Einstein (November 1915): general relativity yields the missing 43 arc-seconds "on the nose." The physical sources of the correction: gravitational time dilation, non-linearities in the space-time metric produced by the Sun's mass, and gravitationally induced curvature of space near the Sun.

Newtonian Theory as a Limit-Case Idealization. Einstein's theory yields Newton's as the static, weak-field limit. This is an entirely different kind of idealization from Newtonian idealizations: a limit-case idealization would never hold exactly in any relevant circumstances; it always holds only approximately, though to asymptotically higher approximation as the limit is approached.

- *Continuity of evidence:* All existing evidence for Newton's theory immediately becomes evidence for Einstein's (though not conversely). Gravity research did not have to restart from the 18th century.
- *Non-nullification:* Einstein's theory does not entail that the purported evidence for Newtonian gravity was never really evidence at all, but only a misleading coincidence. The counterfactual conditionals derived from Newtonian theory remain valid (with qualifications about precision and domain).
- *Conservation of details:* All the configurational details that Newtonian theory revealed as making a difference remained in place after Einstein. Neptune is still the source of the Uranus anomaly; the relative positions of Jupiter's and Saturn's orbit centers still govern the Great Inequality.
- Paradoxically, the 43 arc-seconds turned out to be at once evidence *against* Newtonian theory and evidence *for* it over a restricted domain: Einstein's success further validated the Newtonian derivation of the other 531 arc-seconds of planetary perturbations on Mercury's perihelion.

V. Complications and Sources of Difficulty

Smith identifies three main complications that obscured the logic of theory-testing in the first two centuries of gravity research:

- **Underdetermination of physical sources.** Second-order phenomena generally do not uniquely determine their physical source. (The Uranus anomaly could not determine both Neptune's mass and distance.) When this happens, hypothesis-testing becomes the norm, reinforcing the tendency to see all theory-testing as hypothetico-deductive.
- **Confirming robustness.** The demand is for evidence for the proposed source *beyond* the second-order phenomenon itself. This can require substantial further research: evidence may take the form of hypothetico-deductive predictions, bootstrapping (Glymour), common-origin reasoning (Janssen), convergence of independent measurements (Harper), or evidence from the success of research predicated on the source.
- **Isolating genuine second-order phenomena.** Sources of discrepancies include not only physically significant forces but also: inadequate mathematical approximation (truncated infinite series),

imprecise values of masses, orbital elements, and fundamental constants, and various sources of systematic observational error. (Table 2 of the essay catalogs these.) Spurious second-order phenomena — like the anomaly in Uranus that disappeared when Standish used the correct mass for Neptune (1993) — must be distinguished from genuine ones.

VI. "Evidence for What?" (Section 3.2)

Having laid out the structure of the evidence, Smith turns to the question of what exactly this evidence is *for*. He considers several candidate answers, subjecting each to important qualifications.

Preliminary point: theory-neutrality of observation. Smith emphasizes that before the advent of general relativity, none of the astronomical observations used in comparisons with calculation presupposed Newton's theory of gravity. The observations were corrected for parallax, atmospheric refraction, aberration of light, etc., and reduced using aspects of Keplerian orbital theory — but all of this was independent of the theory of gravity. (This ceased to be true after Einstein: observations now require relativistic corrections that depend on what the theory says about gravity.) This matters because it means that any circularity in the evidential reasoning is *not* a circularity in which the observations themselves are contaminated by the theory being tested.

Candidate 1: Evidence for the accuracy of Newtonian theory. The most immediate bearing of the evidence was on how accurately Newtonian theory holds. Example: the exponent of r in the law of gravity. Newton could only say it deviates from -2 by at most 2 parts in 243; by 1903, Brown had reduced this to at most 4 parts in 10^8. More generally, the 1–2 arc-second agreement between calculated and observed planet locations achieved by Newcomb and Hill confirmed the very high precision of the theory.

Caveat: the evidence confirmed the accuracy only of *those aspects of Newtonian theory that entered indispensably into the calculations*. Smith points out that one important element — the claim that the gravitational field surrounding each body is proportional to its *mass* — made no detectable difference in the orbital calculations before the 20th century. This is because the masses of celestial bodies were inferred from the inverse-square centripetal acceleration fields surrounding them, and entered the calculations only as GM, i.e. as the very acceleration fields from which they were inferred. The masses served as mere *placeholders*. In Smith's words: "before the twentieth century the evidence from orbital motions did not confirm that the mass of the attracting body even has to be present in the numerator of the law of gravity."

More precisely: what actually entered the calculations were various *principles deduced from* the law of gravity that concern specifics of orbital motions — principles that are less general than the law itself. The evidence confirmed the accuracy of *those principles*, and the bounds of precision varied from principle to principle. This matters because these individual principles are what underpin the counterfactual claims about the differences any configurational detail makes.

Candidate 2: Evidence for the projectibility of Newtonian theory. "Projectibility" here (in Goodman's sense) involves two claims: (1) the principles apply not just to actual situations but to a range of "possible" variants; and (2) the parameters entering into those principles, *and no other parameters*, delineate the kinds of changes that can make a difference. Three aspects of the evidence supported projectibility:

- Emergence of *further* discrepancies with clear signatures (especially ones that would have been masked had earlier sources not been incorporated) — evidence that prior clear signatures were not mere coincidences.
- Successes in identifying robust physical sources — evidence that the tighter agreement with observation represents something more than a curve-fit; the parameters involved gained increasing claim to being ones that physically make a difference.

- The overall sequence of successive approximations, addressing a variety of discrepancies involving different orbiting bodies with a single unifying theory, *without having to backtrack* — evidence that the principles and parameters can characterize the full range of physically possible variants.

Post-Einstein qualification: the domain over which the Newtonian principles are projectible is now sharply circumscribed. They remain projectible, but only over a domain defined by the level of precision of the observations. (An alternative formulation: the Newtonian principles amount to *surrogates* for corresponding Einsteinian principles that are not (yet) subject to the same constraints on projection. Viewed as surrogates, they remain projectible because they approximate the Einsteinian principles over a restricted domain.)

Candidate 3: Evidence for the physical sources themselves. The evidence strongly confirmed that specific factors identified as sources of second-order phenomena really are their sources (or close proxies). But what does "being the source" mean? Smith clarifies: to say that certain physical factors are the source of a phenomenon means that those factors made the difference between this phenomenon occurring and its not occurring (or at least occurring very differently). This amounts to asserting a *counterfactual conditional*: if the values of the parameters characterizing those factors had differed notably, then the phenomenon would not have occurred as it did. Evidence for physical sources is therefore evidence for certain counterfactual conditionals — or equivalently, for the comparatively specific principles derived from the law of gravity that lie behind those counterfactuals.

Post-Einstein consequences: After the switch, we also have *lower* bounds on the inaccuracy of Newton's theory (varying locally with GM/r^2c^2), but within the solar system these bounds are generally well below the level of observational accuracy. In some places the evidence for Newtonian claims became *stronger* with the advent of general relativity — e.g., Einstein's success with the perihelia of all four inner planets strengthened claims about the accuracy of Newtonian perturbational effects. The transition had the *least* effect on evidence for physical sources: the counterfactuals describing what difference each detail makes remain in place, equally well supported by the Newtonian principles and their Einsteinian counterparts.

VII. "Knowledge Achieved" (Section 3.3): Tempting Proposals and Why They Fail

Smith considers — and rejects — what he takes to be the most natural proposal about the knowledge achieved in gravity research.

The natural proposal (and its failure). Since the physical sources of Newtonian second-order phenomena turned out to be more resilient than Newton's theory itself (they all survived the transition to Einstein), one might naturally suggest: *those physical sources have all along had more of a claim to being knowledge than the theory that enabled them to be identified.* The details that make a difference — Neptune, the eccentricities governing the Great Inequality, the fluctuation in the Earth's rotation — are the real knowledge; the theory was just the scaffolding.

Smith spells this out carefully, then rejects it: "The trouble is, this proposal, as it stands, makes no sense." The reason:

> Conclusions that some detail makes some specific difference in motions in our solar system cannot be divorced from the theory that tells us it makes this difference. This is especially true in this case insofar as the research yielding such conclusions had no means of intervening to manipulate the details in order to see what happens.

There were no experiments in orbital research (before spaceflight) — no way to, e.g., remove Neptune and see what happens to Uranus. Any claim to knowledge that the details have must accordingly include a claim to knowledge, in some form or other, for the theoretical principles that dictate what differences those details are supposed to be making.

The corrected view: interpenetration of theory and details. Knowledge is better described as an *interpenetration* of theory and the details that make a difference. The details have provided continuing evidence bearing on the theory and increasingly precise values for its many parameters. The theory has been indispensable for identifying which details make a difference and what differences they make, providing the projectible principles underlying the counterfactual conditionals.

What is required of the theory? Smith emphasizes that the claim to knowledge does *not* require Newton's law (or Einstein's field equations) to be the final word. The theoretical principles do not have to be exceptionless generalizations that hold universally. They only have to hold to an *appropriate level of approximation* across a *restricted domain* that includes the actual situations of concern and enough of a range of variants to give the relevant counterfactuals their content. The required level of approximation is dictated by the level of precision assumed in describing the differences the details are claimed to make.

The question of permanence. Will future theories of gravity invariably retain the conservative relationship to Newtonian theory that Einstein required of general relativity? The evidence from the first two hundred years gave Einstein overwhelming reasons to impose a limit-case constraint; the subsequent century has given future researchers even stronger reasons for imposing it. But Smith is a fallibilist "I, for one, do not see how any empirical evidence could ever" eliminate all possibility of future theories nullifying Newtonian conclusions about which details make a difference.

A Duhemian point, reconsidered. the evidence for Newton's theory did not eliminate the alternative theory of general relativity; nor will the evidence for Einstein's theory ever eliminate all logically possible alternatives. But on Smith's view *this Duhemian claim is not of much consequence*, so long as the alternative theories that remain are ones that *conserve* the Newtonian (and Einsteinian) conclusions about which details make a difference. The real question — whether the evidence can eliminate all theories that *do not* conserve those conclusions — Smith again answers in the negative.

Burden of proof. But consider what the evidence demands of any theory that *repudiates* the Newtonian conclusions. Such a theory would have to provide grounds for saying that (a) the degree of agreement between Newtonian calculations and observation over three centuries has been a mere accident, a subtly masked form of curve-fitting; and (b) the *nested sequences* of second-order phenomena — including each case where a discrepancy with a clear signature emerges only after sources for preceding discrepancies have been incorporated — are mere coincidence. This burden is enormously larger than that on theories that conserve the conclusions. According to conserving theories, neither the overall agreement nor the nested sequences have been mere coincidence at all.

VIII. Connections and Discussion Questions

1. **Smith vs. Duhem:** Smith directly challenges the hypothetico-deductive picture Duhem endorsed. But both begin from the same evidence problem (the complexity of true motions guarantees that multiple theories can agree with observation to any given approximation). How adequate is Smith's response? Does the closing-the-loop logic really escape Duhem's concerns about the underdetermination of theory by evidence, or does it merely raise the bar?

2. **Second-order phenomena and theory-mediated measurement:** Second-order phenomena are theory-dependent in exactly the way that theory-mediated measurements are: they presuppose the theory of gravity, taken as exact. How does Smith's account relate to van Fraassen's and Chang's discussions of theory-mediation and circularity? How should we compare Smith's "closing the loop" and Chang's "epistemic iteration"?

3. **Continuity of evidence and theory change:** Smith argues that the transition from Newton to Einstein exhibits "continuity of evidence" — all prior evidence carried over. Is this an instance of a

general pattern in science, or is gravity research atypical? Smith himself cautions against generalizing. What features of gravity research made this possible? (Compare Smith's explicit acknowledgment that the evidence was "misleadingly parochial" — the phenomena of the solar system happened to involve weak, nearly static gravitational fields.)

4. **What makes evidence strong?** Smith claims that the evidence from gravity research was "much stronger than it would have been had the sole test been agreement between calculations and observations." What exactly is doing the epistemic work here? Is it the convergence of independent constraints? The nested, iterative structure? The fact that small errors would propagate and destroy the clear signatures? How do we characterize and compare the "strength" of evidence across different episodes?